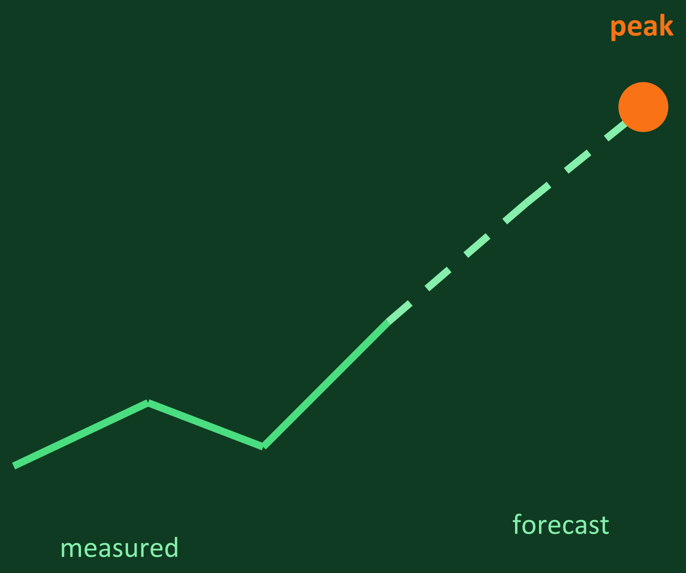
AirSense

Predicting the evening air-quality peak before it arrives.

25.8% better than the naive baseline · R^2 0.801



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The air goes bad before anyone knows

Delhi NCR — and specifically our own campus, which we audited on foot.

Government monitoring stations report city-wide conditions, and they report them after the fact. On a campus that means the decision people actually face — should this session move indoors, should the gate queue be staggered — is always made too late.

1. **Students**
   Outdoor sessions, sport and assembly sit in the evening peak.
2. **Staff**
   No advance signal to reschedule activity or raise filtration.
3. **Commuters**
   The gate and parking areas are the most traffic-exposed points on site.

MEASURED IN OUR OWN DATA

3.3×

higher NOx in the 6–9 PM window than overnight

175.1 ppb evening mean (18:00–21:00)

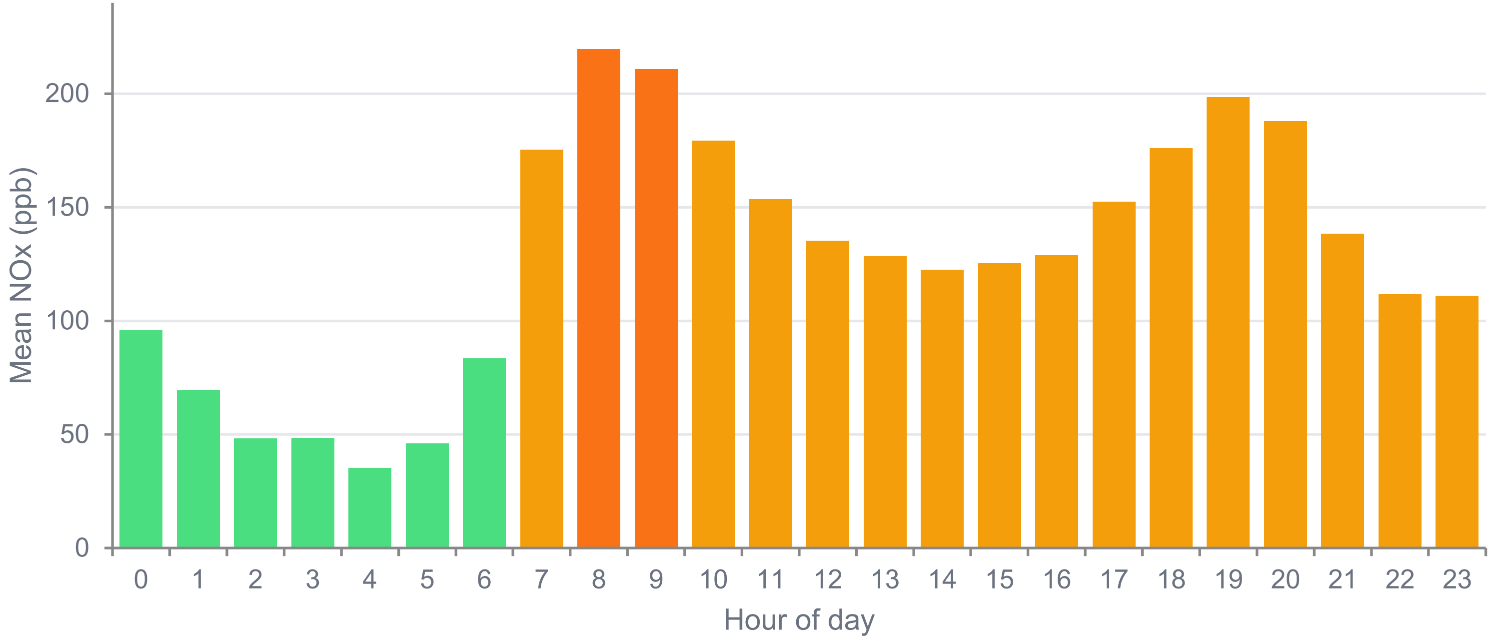
53.3 ppb overnight mean (03:00–06:00)

Computed live from 2,000 hourly readings — never hardcoded.

Pollution here is not random noise. It is a daily cycle that tracks traffic — which is exactly what makes it forecastable.

What the data actually says

UCI Air Quality Data Set (De Vito et al., 2008) — 2,000 contiguous hourly readings.



Bars use the same risk bands as the app: green below 100 ppb, amber 100–200, orange above 200. Two peaks are visible — the morning rush and the 6–9 PM window (hours 18–21).

Evening peak averages 3.3× overnight levels — 175.1 ppb versus 53.3 ppb.

HOW WE PREPARED IT

- Replaced the −200 sensor-failure sentinel with null
- Dropped rows with no target value
- Forward-fill, then median-fill, remaining gaps
- Removed duplicate timestamps
- Sorted chronologically

This mirror was already cleaned upstream, so the sentinel pass replaced 0 values. We report that rather than claim credit for it.

11 Mar – 23 Jun 2004 · hourly · 63 non-hourly gaps noted

How AirSense works

Time-series regression: continuous, timestamped data with a strong daily cycle.

Data → Cleaning → Features → Model → Forecast → Action

1. **Gradient boosting, not an LSTM**

   About 2,000 rows of tabular lagged features. It trains in seconds and its feature importances are directly inspectable — a deliberate trade-off, not a shortcut.

2. **Every feature is lagged**

   CO and NOx correlate at 0.94 here. Using same-hour co-pollutants would be nowcasting dressed as forecasting, so weather and co-pollutants are lagged by one hour.

3. **One shared feature module**

   app/features.py is imported by both training and serving, so the two cannot silently drift apart — the classic way a forecasting project breaks.

BUILT WITH

Python · scikit-learn · pandas

FastAPI · Uvicorn

Chart.js · Tailwind CSS

Docker · Google Cloud Run

Claude Code (AI assistance)

One Cloud Run service serves the API and the UI.

Prediction target: NOx concentration, 1 to 6 hours ahead, with a recommended action attached to every hour.

The live prototype

Deployed and public — no login required.

airsense-526660427489.asia-south1.run.app

!

Peak predicted in 2 hours — act now.

Forecast reaches 239.7 ppb (High) at 08:00. Move outdoor sessions indoors.

WHAT A USER DOES

Opens the link, reads the current band and the action for right now, then checks the six forecast cards for the next window worth acting on.

Dashboard

Current reading, risk band, recommended action, and the six-hour forecast strip.

Forecast

Measured 48 hours joined to the prediction, hour-by-hour table, CSV export.

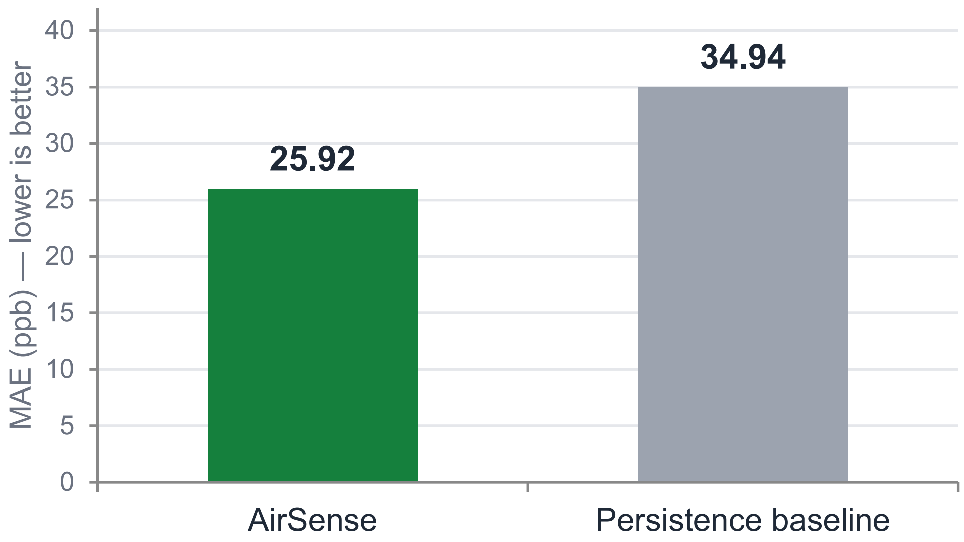
Model Performance

Backtest of predicted against actual, plus early-warning skill.

Because the dataset is historical, "now" is a position you choose — a judge can move the replay point and watch the model re-forecast from only the data before that hour.

Validated, not asserted

Held-out chronological test set — 396 hours the model never saw in training.



Chronological split — no future data leaked into training. Baseline is "persistence": next hour equals current hour, the honest naive benchmark.

25.8%
lower error than baseline

0.801
R^2 on held-out test set

1,580 / 396
train / test rows

86.4%
hours within 50 ppb

EARLY WARNING · ONE HOUR AHEAD

Of 72 hours that actually exceeded 200 ppb, the model flagged **52** — 72.2% caught, 77.6% of warnings correct. 20 missed, 15 false alarms.

Known limits: one sensor stream, so the four campus zones are simulated and labelled as such · historical data, not a live feed · exogenous features held constant · risk bands are project-defined, not CPCB standards.

Impact potential

What changes when a campus can see the peak an hour or more before it lands.

1 Environmental

- Targeted filtration and ventilation instead of running it blind all day.
- Advance notice supports staggering exit timings to flatten the gate-queue peak.
- Measured capability: 72.2% of High-pollution hours flagged one hour ahead.

2 Social

- Sensitive and asthmatic individuals get a decision, not just a number.
- Outdoor sport and assembly can be scheduled into cleaner windows.
- The zone design lets guidance be location-specific rather than campus-wide.

3 Governance

- Every figure in the interface is computed from real data; anything derived is labelled simulated.
- Validation, limitations and data provenance are published in the repository.
- Forecasts export to CSV with provenance headers for audit.

Who benefits first: roughly the whole campus population passing the Main Gate and Parking Block, the two most traffic-exposed points in our design. Quantified ESG targets are carried over from Assignment 4A.

What's next

The single biggest accuracy win available to us is a weather forecast — it is the one assumption we currently hold constant.

1. **Pilot** — Deploy a real low-cost NOx sensor at MAIT and replace the historical file with a live feed.
2. **Multi-zone** — Additional sensors turn the four simulated zones into four measured ones.
3. **Forecast quality** — Ingest a weather forecast so exogenous features stop being held constant.
4. **Alerting** — Opt-in notifications ahead of predicted Severe hours.
5. **Scale** — Multi-campus tenancy and a documented public API.

SUPPORT NEEDED

- A calibrated reference-grade sensor for validation
- Permission for on-campus sensor placement
- Cloud credits beyond the trial tier

Live app · **airsense-526660427489.asia-south1.run.app**

Code · github.com/Rex123-hash/Air-Sense